

DEVELOPMENT OF VISIBLE-LIGHT REDUCTIVE CYCLIZATION REACTIONS FROM UNACTIVATED CHLOROALKANES

Miguel Claros,^a Felix Ungeheuer,^a Alicia Casitas,^{a*} Julio Lloret-Fillol^{a,b*}

^a Institute of Chemical Research of Catalonia (ICIQ), The Barcelona Institute of Science and Technology (BIST),
Avda. Països Catalans, 16., Tarragona, Spain

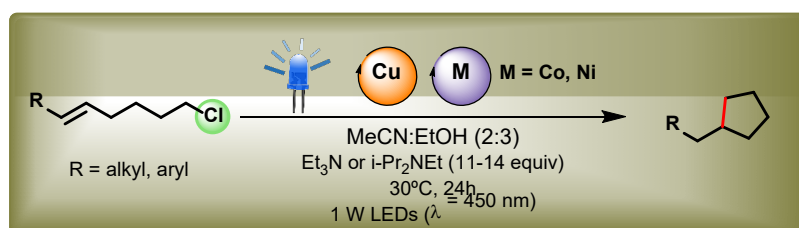
^b Catalan Institution for Research and Advances Studies (ICREA), Paseig Lluís Companys, 26, Barcelona, Spain.

e-mail: mclaros@iciq.es



The state-of-the-art Ru^{II} and Ir^{III} redox-active coordination complexes have enabled the generation of C-center radicals under visible light irradiation. These novel reactivity has opened a more sustainable venue for the formation of C-C and C-heteroatom bonds in highly functionalized molecules. These photosensitizers engage into single electron transfer (SET) reactions towards several electrophilic substrates containing activated or weak C-halogen bonds, mostly aryl and alkyl iodides or activated alkyl bromides. Further efforts into finding alternative photoredox catalysts have opened the redox-potential window of substrates that can be activated, achieving the cleavage of Csp³-Br of unactivated alkyl bromides.

Chloroalkanes are predominant patterns in both commercially available building blocks and biologically active molecules, including pharmaceutical and agrochemical compounds. However, alkyl chlorides present large negative reduction potentials, which render them unsuitable partners for engaging into SET processes from current photoredox catalysts. Taking into account these premises, our goal is to develop earth-abundant transition metal catalysts to cleave strong Csp³-Cl bonds under mild conditions and using visible light as a driving force.



- 1) Curran, D. P. In *Comprehensive Organic Synthesis*; Trost, B. M., Fleming, I., Semmelhack, M. F., Eds.; Pergamon: Oxford, 1991; Vol. 4, pp 715_779.
- 2) Studer, A.; Amrein, S. *Angew. Chem., Int. Ed.* **2000**, 39, 3080.
- 3) Parsons, A. *Chem. Br.* **2002**, 38, 42.
- 4) Tellis, J. C.; Primer, D. N.; Molander, G. A. *Science* **2014**, 345, 433.
- 5) Zuo, Z.; Ahneman, D. T.; Chu, L.; Jack A. Terrett; Doyle, A. G.; MacMillan, D. W. C. *Science* **2014**, 345, 437
- 6) Shaw, M. H.; Shurtleff, V. W.; Terrett, J. A.; Cuthbertson, J. D.; MacMillan, D. W. C. *Science* **2016**, 352, 130